Simplified Multiple-Impulse Spacecraft Rendezvous Optimization

Mini Project Outline

Summer 2026

Disclaimer: this document was majority machine-generated. I have made some minor edits here and there. I have vetted the concept for feasibility.

Introduction and Motivation

This project aims to be a primer for a future course in spaceflight, using a simplified approach to orbital mechanics to allow for a simulation-focused treatment. **You will build simulations and experiment with them**, ultimately producing an orbital rendezvous mission. The goal is to develop intuition by seeing how trajectories evolve and how small changes affect motion.

This project is framed around the following research question:

> *How can a spacecraft use a small number of engine burns to rendezvous (meet up) with another spacecraft orbiting Earth?*

This problem is both practically important and conceptually rich. Unlike simple orbit transfers, rendezvous requires careful coordination of:

- Position;
- Velocity;
- Timing.

By the end of the project, you will have built a program that:

- Simulates spacecraft motion under gravity;
- Applies impulsive velocity changes ("burns");
- Models the motion of two spacecraft simultaneously;
- Searches for sequences of burns that achieve rendezvous.

The emphasis throughout is on:

- Clear physical understanding;
- Computational experimentation;
- Building working tools step by step.

Completion of this project will serve as preparation for courses such as AER306 and AER506, but also computationally-focused academic research projects.

Physics Model

We model motion in two dimensions using Newton's law of gravity:

$$\ddot{\mathbf{r}} = -\mu \frac{\mathbf{r}}{|\mathbf{r}|^3} \tag{1}$$

where:

- $\mathbf{r} = \begin{bmatrix} x \\ y \end{bmatrix}$ is position;
- μ is a constant representing Earth's gravitational strength[1].

Each spacecraft has a state:

$$(x,\, y,\, v_x,\, v_y)$$

You will simulate how this state evolves over time using a numerical solver, similarly to the second order ODEs you implemented in ESC103.

Project Overview (12 Weeks)

Weeks 1–2: Simulating Motion Under Gravity

Goal: Build a basic orbit simulator.

- Implement the equations of motion;
- Use a numerical ODE solver to propagate trajectories;
- Plot the trajectory in 2D space, i.e. $(x,\, y)$, and in time, i.e. $x(t)$, $y(t)$.

Milestone:

- A working simulation showing a spacecraft orbiting Earth.

Weeks 3–4: Exploring Orbit Behavior

Goal: Develop intuition for different types of motion.

- Vary initial velocity and observe outcomes;
- Identify:
  - Circular motion;
  - Elliptical motion;
  - Escape trajectories.
- Plot trajectories and describe observations;

Milestone:

- A short write-up describing different orbit types.

[1] In real life, this constant has a value of $\mu = 3.986 \times 10^{14} \mathrm{m}^3/\mathrm{s}^2$, equal to the mass of Earth times Newton's universal gravitational constant. For more info, read this Wikipedia page.

Weeks 5–6: Impulsive Maneuvers

Goal: Model spacecraft engine burns.

- Implement instantaneous velocity changes:

$$\mathbf{v} \to \mathbf{v} + \Delta\mathbf{v}$$

- Apply burns at specified times;
- Observe how trajectories change.

Milestone:

- Simulation of a single burn altering an orbit.

Weeks 7–8: Multiple Burns and Target Spacecraft

Goal: Simulate two spacecraft.

- Add a second spacecraft (the "target");
- Propagate both under the same gravity model;
- Implement multiple burns for the "chaser";
- Plot both trajectories together.

New concept:

- Relative distance:

$$d(t) = |\mathbf{r}_c(t) - \mathbf{r}_t(t)|$$

Milestone:

- Visualization of two spacecraft and their separation over time.

Weeks 9–10: Rendezvous as an Optimization Problem

Goal: Define and evaluate "how close" you get to rendezvous.

At final time T, define:

- Position error:

$$J_r = |\mathbf{r}_c(T) - \mathbf{r}_t(T)|^2$$

;

- Velocity error:

$$J_v = |\mathbf{v}_c(T) - \mathbf{v}_t(T)|^2$$

;

Total cost:

$$J = w_r J_r + w_v J_v + w_{\Delta v} \sum_i |\Delta\mathbf{v}_i|$$

.

Tasks:

- Try different burn strategies;
- Evaluate the cost for each.

Milestone:

- A function that scores how good a trajectory is.

Weeks 11–12: Searching for Good Trajectories

Goal: Automatically find better solutions.

- Programatically try many combinations of:
  - Burn times;
  - Burn directions;
  - Burn magnitudes.
- Use simple strategies:
  - Grid search;
  - Random search.

Milestone:

- A program that finds a sequence of burns that approximately achieves rendezvous.

Final Deliverable

A program that:

- Takes:
  - Initial spacecraft state;
  - Target spacecraft state;
  - Number of burns.
- Outputs:
  - A sequence of burns (time, magnitude, direction);
  - A trajectory plot showing rendezvous attempt.

General Advice

- Focus on building something that works, not something perfect;
- Use plots to understand what is happening;
- Change one parameter at a time when experimenting;
- Expect things to fail — this is part of the process!